

WordPipe Field Notes

The gradient-free specialist-genome language pipeline (/evolang)

Complete findings record, archived 2026-07-08, before a ground-up rebuild.

GENREG project — `genreg_train/wordpipe*.py` and dependents

Executive Summary

WordPipe is the thesis that a language model can be built entirely gradient-free: not one network trained on a loss, but a pipeline of tiny evolved specialist genomes, each with a single falsifiable job, composed. The core architectural principle (the "GA abstraction thesis"): evolution does not discover an embedding space by searching it — that space is too large. Instead the space is BUILT from corpus statistics (induced word classes via k-means, SVD distributional features, mined relation pairs), and evolution learns one tiny relationship inside that pre-built space at a time. The features are the environment; evolution is the organism navigating it.

Over three corpus generations (Gutenberg novels → Wikipedia → Wikipedia + Cornell Movie Dialogs) and roughly 45 distinct genomes attempted, the pipeline reliably solved local, structural, single-word-pair problems (grammar rhythm, agreement, adjacent-word semantic fit, sentence-boundary placement) but never solved GLOBAL sentence coherence or clause completeness. That is the central, honest finding of this entire body of work, and it is the reason for archiving: the current architecture has been pushed hard on every axis available to it (better vocabulary, decomposed observability, meaning-first ordering, punctuation-first generation, backward growth, stateful obligation tracking) and the fluency ceiling has not moved. Whatever comes next needs a structurally different idea, not another genome bolted onto the same skeleton.

What survived, in one sentence: 13 shipped genomes (vocabulary, class-order skeleton, bidirectional word selection, boundary/comma placement, agreement, alternation, semantic adjacency, no-repeat, opener, closer, chunk lookup) plus a validated-but-experimental layer (sentence-type, pronominalization, three relation genomes, a full structural decomposition for observability, and a working intent-first/backward-generation mode) — none of it adds up to a sentence a person would write.

1. Architecture & Core Thesis

1.1 The pipeline stages

Generation was organized into pipeline STAGES, each a real point in the data flow: **Skeleton** (Order genome — next grammatical class from the last 4 classes), **Fill** (word selection into each class slot), **Boundary** (sentence/comma punctuation), plus three later, larger additions: **Content** (meaning-first — pick content words before structure), **Intent** (punctuation-first — pick the discourse skeleton before any words, grow backward), **Revision** (best-of-N whole-sentence scoring, post-hoc), and a partially-built **Passage** stage (cross-sentence, only pronominalization landed).

1.2 The layer taxonomy

Genomes were also tagged by LAYER — a metaphorical, not code-enforced, grouping: **Structural** (form: grammar, position, rhythm — never touches meaning), **Semantic** (meaning: distributional word-feature spaces built from corpus statistics), **Abstraction** (relations composed on top of the semantic space — the newest, least-tested tier). A genuine architectural lesson surfaced repeatedly: many attempted genomes asked a LAYER-3 question ("do these two relations agree?") using LAYER-2 machinery (raw distributional offsets) — Analogy is the clean example, chance-level no matter how long it trained, because A:B::C:D is a relationship between the OUTPUTS of relation genomes (hypernym, meronym, synonym/antonym), not a pattern in raw SVD offset vectors.

1.3 GA training machinery (shared, reused throughout)

- **BilinearPop / train_pairwise** — genome = a small bilinear head M ; $\text{score}(a,b) = f(a) \blacksquare M f(b)$; trained as a corrupted-pair discriminator against hard negatives.
- **WindowPop / train_windowed** — genome = class embedding + one-hidden-layer validity net over a K -class window; tells a real window from one corrupted at a chosen slot.
- **UnaryPop / train_unary** — genome = a linear scorer over per-word features; used for positional questions (is this word a plausible opener/closer/question-starter).
- **OrderPop / run_class_lm** — genome = embedding + positional weight + tanh hidden layer + softmax; a tiny autoregressive class-sequence (or, reused this session, punctuation-sequence) predictor.
- All trained by the same tournament-selection + elitism + self-adaptive-mutation + energy-homeostasis GA step (wp.ga_step) — no gradients anywhere in the model.

2. Corpus History — Three Generations

Corpus	Size	Why swapped	What it fixed / didn't
Gutenberg novels (project/EEC-main/ engine/corpus.txt)	49MB	Original corpus.	Baseline. 19th-century vocabulary ("thou", "shalt") baked into every generated sample.
Wikipedia (corpora/wikipedia/ wiki_corpus.txt)	316MB	"Modern corpus" directive — kill the archaic vocabulary.	Fixed vocabulary completely (zero archaic words in samples). Did NOT fix fluency — grammar was at least as broken, arguably worse (heavy "of the X of the Y" chains). Also revealed Wikipedia's register barely uses exclamation marks the way emotional language does — every probe word (wow/amazing/hooray) was out-of-vocabulary.
Combined (corpora/combined/ combined_corpus.txt)	421MB (24.4% dialogue)	Wikipedia + Cornell Movie Dialogs (17.1MB real dialogue, repeated 6x for real influence) — real questions, real exclamations, real turn-taking.	Exclaim-affinity training samples rose 2,499 → 133,321 (53x). Corpus exclaim-rate in mark mining rose 0.11% → 2.34%; question-rate 0.07% → 6.69%. Modern/dialogue-specific words ("youtube", "gimme") now genuinely appear in output. Fluency STILL not fixed — this was explicitly not attempted as a fluency fix, only a vocabulary/intent-signal fix, paired with the intent-first architecture change (below) so it wasn't a repeat of the Wikipedia mistake.

Lesson, stated by the user and confirmed by data: "this is a living model, changing one thing isn't going to magically fix everything." A corpus swap changes vocabulary and available signal; it does not change what the Order/Selection genomes structurally can or cannot represent. Every corpus swap in this project's history fixed exactly what it targeted and nothing else.

3. Genome Inventory

3.1 Shipped (13 genomes, default ON)

~9,554 params + chunk lookup. Deploy ~140-450 KB. All PASS on the original generation-time battery (targeted metric + adj-hit/distinct guardrails).

Genome	One thing	Result
Vocabulary	Emit real English words	19%→52% valid
Order	Next grammatical class from last 4 classes	ppl 10.91, 75% of ceiling
Selection	Word that fits the previous word	+15 pts adj-hit
Bidirectional Selection	...and the next class too	adj 0.777, distinct 0.231
Boundary	Where sentences end	learned 18.6-word rhythm
Commas	Where commas go inside sentences	beats base rate
Agreement	Subject/verb, modal/aux agreement	12/12 minimal pairs
Alternation	Content/function word rhythm	func-func pairs -34%
Semantic (adjacency)	Adjacent content words co-occur	content-adjacency 33%→40%
No-repeat	Don't reuse a recent content word	repetition 2.2%→0%
Opener	First word is a plausible starter	bad-openings 17%→8%
Closer	Sentences end on noun/verb, not "of/the"	bad-endings 65%→33%, rate-preserving
Chunks	Emit whole real phrases on class match	+7.5 pts adj-hit

3.2 Experimental (wired, OFF by default)

"Tested" for round-1 genomes originally meant probe + a light spot-check (20-30 samples, 1-2 metrics) — NOT the full guardrail battery. When the full battery (adj-hit, distinct, dangling-rate, mean-length, 60 samples) was finally run, EVERY ONE of the round-1 experimental genomes regressed at least one guardrail. None graduated to shipped. This was an important process lesson: a spot-check systematically under-catches side effects.

Genome	Toggle	Full-battery result
Hypernym / Meronym / Synonym-Antonym	hyper/mero/syn ant	10/10, 9/10, 11/14 probes respectively — standalone-valid, never battery-tested in generation.
Sentence type	sent_type	Dangling-ending rate 20.8%→24.8%. Real question-rate effect confirmed, real fluency cost.
Sentence length plan	lenplan	Dangling-ending rate 20.8%→26.0% — worse than sent_type. (Corrects an earlier "practically inert" call that only checked sentence length, which didn't move, and missed this.)
Pronominalization	pronominal	Dangling-ending rate 20.8%→24.0%. "it"-substitution effect real (0.77%→1.11% of words) but has a fluency cost the earlier spot-check didn't catch.
Revision (Best-of-N)	generate_revision()	Mean sentence length collapsed 14.7→7.8 words (-47%) — the whole-sentence scorer's penalties accumulate with word count, biasing toward short "safe" sentences.

3.3 Cut, with the specific reason (selected — the full list runs to ~30 genomes)

Genome	Why cut
Clause template / Verb argument / Pronoun reference	"Class-level redundancy wall" — Order's 4-class context already subsumes any class-level windowed-validity signal. No headroom: baseline already emits ~0.2% rare class-trigrams.
Preposition completion / Determiner binding	Redundant with Alternation for the illegal cases; determiner-binding had a strong signal (0.776) and real effect (orphan-det 11→0) but adj-hit dropped at EVERY gamma — 14 function-type features too coarse to pick the RIGHT noun.
Tense consistency	Diffuse signal, subsumed by Agreement's local finiteness (54.9→56.4, no real effect).
Sentiment (monolithic)	Mining inverted — "war" scored +7.85, "joy" scored -7.38. Seed-propagation drifted onto generic frequent words, not true polarity.
Polysemy	val_acc looked strong (0.88) but measured the wrong thing — Wikipedia's register skews classically polysemous words to one dominant sense, defeating the neighbor-spread proxy.
Analogy	Chance-level (0.48-0.53) — a layer-3 question (do two pairs share a relation?) asked with layer-2 raw-offset machinery. Needs a genome over relation-genome OUTPUTS.
Register	Quote-span mining conflated narration-about-feelings with real dialogue markers; incoherent probes.
Clause count	Spearman correlation between opener score and TRUE empirical compound-rate = -0.009 — genuinely unlearnable from the opening word alone, not a compound question.
Sentence coherence / Theme consistency	val_acc 0.525 / 0.531, barely above chance — mean-pooling several content words' features into one centroid likely destroys the information a linear head needs, not proof coherence itself is unlearnable.
Definiteness	Blocked before training — "a" is not in the 4000-word pipeline vocabulary at all (min_len=2 drops single-letter words); a vocabulary-construction gap, not a learnability gap.
Transitivity	Redundant — collapses onto the already-shipped Closer genome or the already-cut Verb-argument ambiguity; no POS tags to separate the two.
Open-obligation / Clause-obligation trackers	Both a real, correctly-targeted idea (track unclosed prep/det or relative-clause structure, suppress boundary probability) whose implementation cost more than it fixed — flat suppression forced long run-ons; a decay variant showed no differentiation, traced to a suspected bug in the hand-rolled test harness (see §6).

4. Major Architecture Experiments

4.1 Meaning-first generation

User diagnosis: the pipeline was structure-first the whole time — Order picks a class skeleton blind, meaning gets bolted on afterward as a rerank. Flipped: `_select_content()` picks 3-5 mutually-related content words FIRST (via Hypernym/Meronym/Synonym-Antonym/Semantic, stochastic sampling), then the SAME Order/Fill genomes place each reserved word into the first matching class slot instead of running selection there. No new training.

Verified: selected content sets score +0.58 higher on relatedness than random sets (t=8.25, 40 samples). Placement rate 66.5% (a third of chosen words never find a matching slot and get dropped). **Does not fix word-level fluency** — only WHICH words fill content slots changed, not how well the surrounding function words glue them together.

4.2 Structural genome decomposition (full observability)

After four grammar-fix hypotheses failed (see §5), the direction changed from "find the fix" to "build the ability to trace WHY output looks a specific way." Every shipped structural genome was assessed for a genuine internal compound question:

Parent	Split into	Why compound
Selection	Sel-backward / Sel-forward / Sel-frequency	Already 3 separate terms (ML/MR/beta) summed into one score — split needed no retraining.
Order	Order-bigram (K=1) / Order-context (K=4)	Tight local transition vs wider 4-class context. Order-bigram trained (beats unigram baseline) but left unwired — its artifact is a full class-LM, not a bilinear rerank.
Alternation	Altern-rhythm / Altern-func-chain	Coarse content/function rhythm (val_acc 0.53, barely above chance) vs specific function→function legality (0.68, much stronger) — most of Alternation's real power lives in the subtype detail, not the coarse rhythm.
Agreement	Agree-modal / Agree-number	0.75 / 0.68 respectively — two genuinely distinct grammatical phenomena.
Semantic	Sem-adjacent / Sem-window	0.68 / 0.54 — loose topical fit is a genuinely harder signal than tight collocation.

Real, measured generation-time effect (the only experiment this session with a clean multi-metric win): with decomposition active vs. disabled, func-func-adjacency -32%, distinct-ratio +14%, dangling-ending rate -67%. Visually confirmed in the actual text — the "of the X of the Y" pattern was measurably and visibly reduced. Traded for a different, smaller residual tic ("whom"/"including" used oddly as glue words).

4.3 Intent-first generation (punctuation as anchor, backward growth)

User's idea: the punctuation mark IS the intent, chosen before any word exists — grammar grows backward to serve it instead of forward hoping to land on a valid ending. Implemented with zero new algorithm: Order and Selection are generic autoregressive predictors over whatever sequence they're trained on, so training them on the corpus read BACKWARD (a cache-reversal trick — same `wp.run_class_lm/wp.run_biselection`, same word→class mapping) gives real backward Order+Selection for free. `intent_punct.py` (new) is a tiny autoregressive model over the 6-symbol mark alphabet `{. , ; : ! ?}`, mined directly from the corpus with zero labeling. `sent_type_exclaim.py` generalizes the existing question/statement genome to exclaim via a second binary genome (same decompose-into-binary pattern as §4.2).

`Service.generate_intent_first()`: generates the punctuation sequence for the whole response first, then grows each word-span BACKWARD from its mark toward the previous one, with the mark's TYPE biasing what grows toward it.

Verified working, real output, corrected once (see §6): real questions ("who?", "whom?") and a real exclamation ("who!") appeared tied to genuine discourse marks — the first time this session real intent-carrying punctuation appeared in generation at all. Does not fix fluency; sentences remain word salad. This changes WHAT anchors generation, not the underlying word-to-word mechanics.

4.4 Revision stage & crystallize

`generate_revision()`: for each sentence slot, generate several candidates with the unchanged pipeline, score with `_sentence_score()` (composite of already-evolved champions), keep the best. Mechanically verified (correctly rank-orders 8 candidates) but has a severe length bias (mean length -47% in the full battery — see §3.2).

`crystallize` (optional flag on intent-first): a forward polish sweep after the backward pass, re-picking each word against its LEFT neighbor too, using the shipped forward Selection genome. Real result once two implementation bugs were fixed: `func-func-adjacency` -0.026 (better), `dangling-rate` +0.068 (worse). Mixed — stays off by default; no visible readability gain in the actual samples for a second full pass's worth of compute.

5. Grammar / Fluency Investigation — Every Hypothesis Tried

The central open problem: generated text reads as word salad regardless of which genomes are active. Every hypothesis below was tested with real measurements, not assumed.

#	Hypothesis	Verdict
1	Stub-article density in Wikipedia is the cause	Disproven — template phrases ("is a town") appear in under 0.5% of articles; top bigrams look like ordinary English frequency stats, not a Wikipedia artifact.
2	Alternation strength is too low	Disproven, harmful — sweeping gamma up to 6x default made every metric WORSE (func-func-adjacency +19%, distinct -41%, dangling +200%). The chain is the DEGENERATE LIMIT of forced alternation, not a sign of too little.
3	Suppress literal bigram repeats ("of the" recurring)	Narrow win, no real quality gain — cut literal recurrence 62% but func-func-adjacency and distinct barely moved; dangling-rate rose 66%.
4	Selection's frequency-bias term is the culprit	Classic diversity-vs-plausibility trade-off, not a fix — zeroing the frequency term raised distinct +45% but adj-hit collapsed 23 points.
5	Structural decomposition (§4.2)	The one real, clean multi-metric win this session — but addresses function-word chaining, not sentence-level coherence.
6	Clause-obligation tracker, flat gamma	Small real effect (never-closed-relative-rate 13.1%→12.5%, plateaus immediately) at a real cost (dangling worse) — cut.
7	Clause-obligation tracker, decay variant	Inconclusive — all 9 gamma/tau combinations produced byte-identical results, traced to a suspected independent bug in the hand-rolled test harness (110+ word "sentences" even at baseline). Never resolved before archiving — see §7 recommendations.

User's conclusion, confirmed by every data point above: no single knob fixes this, because Order and Selection are both fundamentally LOCAL — next class from the last 4 classes, word score against immediate neighbors — neither has any model of what the sentence has already said. This is the actual reason for archiving: seven independent hypotheses, seven honest non-wins, converging on the same structural ceiling.

6. Process Lessons (worth carrying into the rebuild)

6.1 Spot-checks systematically under-catch side effects

Every round-1 experimental genome passed a lightweight spot-check (1-2 metrics, 20-30 samples) and then FAILED the full guardrail battery (4 metrics, 60 samples) once it was finally run. The fix adopted: run the full battery as a required last step before calling anything validated, not probe + one spot-check metric.

6.2 Silent no-op bugs are the most dangerous class

Three separate incidents this session where code executed without crashing but silently did nothing: (a) self.champs empty on the I2 primary because demo/genomes.pkl is not in the deploy whitelist and a verification script never overrode the cache path — every "X in self.champs" check just quietly evaluated False, and reranks were inert without disclosure; (b) the crystallize forward-polish branch's own guard depended on the same empty-champs condition, so it silently never ran, producing byte-identical A/B output that was initially misreported as "verified, no difference"; (c) the decay-based clause-obligation sweep produced identical results across 9 parameter combinations, later traced to a suspected bug in the underlying hand-rolled generation loop (110+ word "sentences" even at baseline) rather than the decay logic itself. **Lesson: byte-identical A/B results are the tell for a silent no-op, not evidence of "no effect" — always suspicious, always worth a diagnostic counter before trusting.**

6.3 Hand-rolled parallel reimplementations of generate() are a liability

Every standalone experiment script that reimplemented a simplified version of the generation loop (rather than adding a toggle to the shipped, proven Service.generate()) introduced a NEW implementation bug: missing Closer/Opener biases silently changing baseline behavior, cache-path mismatches, list-length misalignment when class sequences skip the reserved class. Recommendation for the rebuild: prefer adding toggles to one proven generation function over spinning up new parallel copies for every experiment.

6.4 Pushing to the I2 primary while a job runs kills it

push_to_primary.py always triggers a node restart, which terminates any in-flight job (by design, to avoid orphaned processes — a deliberate earlier fix). This cost two jobs this session before the discipline was established: confirm the job queue is empty before every push.

6.5 The class-level / word-level redundancy wall (2026-07-07 finding, still true)

The single most load-bearing lesson from the ORIGINAL battery, and the one that predicted most of this session's cuts: genomes that ask a validity question at the CLASS level (clause template, verb argument, pronoun reference) are always subsumed by Order's 4-class context — there's no headroom, Order already knows which class sequences are common. Genomes that ask a validity question at the word/function-type level (determiner-binding, preposition-completion) are always subsumed by Alternation for the same reason. **The genomes that survive are the ones enforcing a constraint Order/Selection/Alternation STRUCTURALLY cannot see** — semantic co-occurrence (content identity, not class) and exact-word recency (state, not a static rule).

7. What to Keep vs. Archive

7.1 Keep — datasets and durable infrastructure

Path	What it is	Why keep it
corpora/wikipedia/wiki_corpus.txt	316MB Wikipedia text dump	Real, cleaned, ready-to-tokenize modern-vocabulary corpus.
corpora/wikipedia/wiki_feats.npz	30K-vocab, 128-dim SVD distributional features	Expensive to rebuild; validated with real nearest-neighbor sanity checks (king→queen/prince/emperor).
corpora/combined/combined_corpus.txt	421MB Wikipedia + Cornell Movie Dialogs blend	The best available corpus for real intent-carrying punctuation; took ~94 min to retrain everything against.
project/conversational/cornell_movie-dialogs_corpus/	Raw Cornell dialogue dataset	Source data for the combined corpus; genuinely modern, real turn-taking.
genreg_train/wordpipe.py	Core GA training machinery (OrderPop, BilinearPop, WindowPop, UnaryPop, ga_step, gen_class_seq, boundary_prob, etc.)	Generic, reusable, well-tested infrastructure independent of any specific genome.
genreg_train/genelib.py	Shared pairwise/windowed/unary training scaffolding	Same — infrastructure, not a specific model.
I2 job-dispatch system (i2_node.py + push_to_primary.py + run_job.py)	Signed, whitelisted remote training dispatch to the I2 primary node	Proven, reusable compute infrastructure independent of WordPipe specifically.

7.2 Archive — the pipeline itself

Everything under the WordPipe genome umbrella: `genreg_train/wordpipe_service.py` (the `Service` class and every `generate*` method), all ~45 individual genome modules (`agreement.py`, `altern.py`, `sem_compat.py`, `sent_type.py`, `intent_punct.py`, and every cut module alongside them), the decomposition modules, the backward-generation experiment scripts, `templates/evolang*.html`, `static/evolang*.js`, and the `/evolang-family` routes in `app.py`. Archive as a snapshot (branch or dated directory), not delete — every cut reason and every real number in this document traces back to a specific file worth having on hand during the rebuild.

8. Recommendations for the Rebuild

- **Start from the architecture, not the genome list.** The single clearest lesson: structure-first generation with meaning/intent bolted on afterward has been tried in every combination (forward, meaning-first, intent-first, backward, crystallized) and none reached sentence-level coherence. Consider whether the class-induction step itself (32 k-means clusters over left/right anchor context) is the real ceiling — it has no notion of subject/object/clause boundary by construction, and every genome that tried to add that notion at the class level got redundancy-walled by Order.
- **Rebuild the guardrail battery FIRST, before any genome ships default-ON again.** The spot-check-then-battery pattern cost real time and one public overclaim this session (§6.1, §6.2). Make the full battery (adj-hit, distinct, dangling-rate, mean-length, 60+ samples) step zero of the genome-acceptance protocol, not an afterthought.
- **One proven generate() function, toggles not forks.** Every hand-rolled parallel reimplementations introduced a new bug (§6.3). If the rebuild keeps a similar composed-genome architecture, invest early in making the real generation function flexible enough that experiments are toggles inside it, not new files.
- **Keep the decomposition discipline.** The full observability directive (§4.2) was this session's one clean, multi-metric win. Whatever architecture comes next, build in traceability (which sub-signal produced which word) from day one rather than retrofitting it after the fact.
- **Resolve the decay clause-obligation harness bug before concluding anything about clause-completeness.** The last open thread (§5, #7) was never actually resolved — the conclusion "neither obligation-tracker mechanism works" rests partly on a suspect test harness. Worth a clean re-test on the new architecture rather than importing the conclusion unverified.
- **Definiteness's vocabulary gap is a real, separate, cheap fix (§3.3)** — raising min_len in the lexicon builder to include "a"/"I" is independent of the architecture question and worth doing early in the rebuild regardless of what replaces the pipeline.

Appendix: Key File Index

File	Role
genomes.txt	The full, continuously-updated running log this document was distilled from — every genome, every verdict, every number, in chronological order.
static/evolang_layers.js	The live flow-map data — source of truth for CURRENT status of every genome (shipped/experimental/cut/deferred), exportable as JSON.
CHANGELOG.md	Dated entries for every change this session, cross-referenced with genomes.txt for the reasoning behind each.
genreg_train/battery_round1.py	The full guardrail battery script (adj-hit, distinct, dangling-rate, mean-length) — needs re-running on any future corpus before trusting a genome's default state.
genreg_train/run_retrain_combined.py	The most recent full-pipeline retrain recipe (23 genomes) — the template for retraining everything after any future corpus change.
corpora/combined/build_combined_corpus.py	How the combined corpus was built — reusable if more dialogue-style data needs mixing in later.